

Introduction

Linear regression rests on four classical assumptions — linearity, independence, homoscedasticity, and Normality of residuals — each with characteristic diagnostics and remedies when violated. Distinguishing the four is essential because their failures have very different consequences: non-linearity biases coefficients, dependence deflates standard errors, heteroscedasticity gives unbiased coefficients with wrong SEs, and Normality matters mainly for small-sample inference. A responsible analysis checks all four and reports the verdict explicitly.

Prerequisites

A working understanding of linear regression, residuals, and the standard regression diagnostics.

Theory

- **Linearity:** $\mathbb{E}[Y | X] = \beta_0 + \sum \beta_j X_j$. Failure produces biased coefficients regardless of sample size.
- **Independence:** residuals uncorrelated with each other (and with predictors). Time-series, repeated-measures, or clustered data violate this; the consequence is downwardly biased standard errors and inflated Type I error.
- **Homoscedasticity:** $\text{Var}(\varepsilon | X) = \sigma^2$ constant across predictor space. Failure produces unbiased coefficients but wrong standard errors; the fix is robust (sandwich) SEs or weighted least squares.
- **Normality of residuals:** needed for exact small-sample t -test and confidence-interval validity. The central limit theorem makes this assumption mild for large n , where coefficient sampling distributions are approximately Normal regardless of residual distribution.

Assumptions

These four are themselves the assumptions. Diagnostics aim to detect violations, not to “prove” the assumptions hold.

R Implementation

```
library(car); library(performance); library(lmtest)

fit <- lm(mpg ~ wt + hp, data = mtcars)

# Residuals-vs-fitted: linearity and homoscedasticity
plot(fit, which = 1)

# Q-Q plot of residuals: normality
plot(fit, which = 2)

# Scale-location: homoscedasticity
plot(fit, which = 3)

# Residuals vs leverage: influence
plot(fit, which = 5)

# Formal tests
bptest(fit)      # Breusch-Pagan for heteroscedasticity
shapiro.test(residuals(fit))
dwtest(fit)      # Durbin-Watson for autocorrelation
```

```
# performance::check_model combines all  
check_model(fit)
```

Output & Results

Four diagnostic plots from `plot.lm()` plus three formal tests: Breusch-Pagan for heteroscedasticity, Shapiro-Wilk for residual Normality, Durbin-Watson for first-order autocorrelation. `performance::check_model()` produces an integrated multi-panel diagnostic with annotations.

Interpretation

A reporting sentence: “Diagnostic plots showed no systematic patterns: residuals-vs-fitted scattered randomly around zero, scale-location was approximately horizontal, and the Q-Q plot showed only mild tail deviation. Formal tests confirmed: Breusch-Pagan $p = 0.41$ (no evidence of heteroscedasticity), Shapiro-Wilk $p = 0.32$ (residuals consistent with Normality), Durbin-Watson 1.95 (no autocorrelation). All four classical assumptions are reasonably supported.” Always report what was checked and the verdict.

Practical Tips

- Graphical diagnostics are more informative than formal tests, especially at small samples; formal tests have low power below $n = 50$ and reject too easily above $n = 1,000$.
- Heteroscedasticity is common; use robust (HC3) SEs from `sandwich::vcovHC` combined with `lmtest::coeftest` if detected. Alternatively, transform the response or use weighted least squares.
- Clustering or repeated measures invalidate the independence assumption; fit a linear mixed model (`lme4::lmer`) rather than ordinary regression.
- Non-linearity is often fixed by a transformation (log, square root), polynomial terms, or splines. The right fix depends on the substantive functional form.
- Gross departures from Normality at large n typically still allow valid CLT-based inference; the assumption matters most for t -tests and CIs at small n .
- Reporting “all assumptions met” without specifying which were checked and how is uninformative; cite the specific diagnostics used.

R Packages Used

Base R `plot.lm()` and `shapiro.test()`; `lmtest::bptest()` and `dwtest()` for heteroscedasticity and autocorrelation; `car::ncvTest()` and `durbinWatsonTest()` for alternatives; `performance::check_model()` for integrated diagnostics; `sandwich` and `lmtest::coeftest()` for robust SE inference.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook’s distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors

- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI